Machinable Boards®

ıe Ren Shape family boards comprises 16 different products for the CNC-machining of dimensionally stable, high :curate models, fixtures, foundry patterns, and tooling. Each product is the line exhibits its own unique combination ıysical characteristics, including varied densities, flexural, tensile and compressive strengths, glass transitic mperatures, and coefficients of thermal expansion. And, all of the Ren Shape boards produce outstanding surfa ıishes and edge definitions.

'YLING/ MODEL BOARDS

120 Styling Board

very low-density (15 pounds per cubic foot) polyurethane board with a Shore 28D hardness. Formulated for machinin
tremely lightweight styling models that exhibit an outstanding surface finish for this low density.

50 Styling Board

economical, lightweight polyurethane board for milling precise styling and sight models, verifying CAD designs ar
aking temporary models and tooling aids. Provides a Shore 30-35D hardness and a good surface finish.

10 Styling Board

low-density polyurethane board for the production of temporary master models, styling models and prototypes, ar
sual models. Exhibits a Shore 58D hardness and produces a smooth surface finish with good edge definition.

140 Modeling Board

moderately low density polyurethane board that allows fast CNC machining and hand working of permanent
mporary master models, as well as styling or visual models. Excellent edge definition.

50 Modeling Board

medium-density polyurethane modeling board that holds tight tolerances, maintains outstanding dimensional stabili
ter exposure to temperature and humidity extremes, and holds solid, well-defined edges.

003 Modeling Board

low-density epoxy board for production of models and low-temperature cure prepreg tools that can withstar
posure to temperatures up to 140°F (60°C). Holds close tolerances well defined edges and exhibits good tensile ar
mpressive strength.

50 Modeling Board

higher-density precision modeling board, the material also exhibits a lower coefficient of thermal expansion tha
her modeling materials for enhanced dimensional stability over a broader temperature range. Exhibits excelle
curacy and edge definition.

OOLING BOARDS

70 Fixture Board

high-density, premium-quality polyurethane board that is designed for CNC-machining production fixtures capable
ithstanding extended use and handling. Maintains its dimensional stability even when exposed to temperature ar
umidity extremes. Also features high compressive and impact strength and good wear resistance. Useful for machinin
andrels for making nickel electroform shells.

'1 Fixture Board

lower-density polyurethane board for machining fixtures produced for light to medium use. Offers good dimensior ability and wear resistance.

'2 Fixture Board

polyurethane board for machining durable checking and assembly fixtures. Combines light weight with goc ughness, flexural strength and abrasion resistance to withstand demanding production environments.

i0 Intermediate-Temperature Board

heat-resistant polyurethane board for forming large, dimensionally accurate models and curing tools that ca ithstand temperatures up to 200°F (93°C). Features good flexural, tensile and compressive strength.

)05 Intermediate-Temperature Board

low-density epoxy board for building 250°F (121°C) master models, prepreg lay-up tools and for other heat-resista plications. Exhibits a low coefficient of thermal expansion for good accuracy and dimensional stability.

)08 Intermediate-Temperature Board

low-density epoxy designed for tooling use up to 250°F (121°C) for building master models and low and mediur mperature curing prepregs. Low coefficient of thermal expansion of 18 x 10-6 in/in/°F produces outstanding accura id dimensional stability

)10 Tooling Board

iis high-temperature, low-density board for molds offers a combination of excellent machinability, and much high ass transition temperature than more conventional tooling boards.

.66 Metalforming Board

tough, durable board for machining dies for drawing, stretch forming and low pressure hydroforming. Can l achined at high feed rates and produces dies with excellent impact and wear resistance, high flexural modulus ar od compressive strength. A companion castable version, RP 6470, is also available.

.68/5169 Foundry Boards

:tremely tough, abrasion-resistant boards available in green (5168) and red (5169) colors for machining found itterns, core boxes and other tooling. Offers good compressive and tensile strength and excellent edge definition ar mensional stability.

odeling & Styling Boards

\pplications	**5120**	**350**	**440**	**5440**	**450**	**5003**	**460**
\IC tape proofing	X	X	X	X	X	X	

Visual models	X	X	X	X	X	X	X
Styling prototypes	X	X	X	X	X	X	X
Architectural models	X	X	X	X	X	X	X
Tooling aids & fixtures							X
Temporary models	X	X	X	X	X	X	X
Master models			X	X	X	X	X
Automotive die models			X	X	X	X	X
Prototype foundry patterns			X	X	X		X
Prototype vacuum-form molds			X	X	X	X	X

Comparison of Benefits	**5120**	**350**	**440**	**5440**	**450**	**5003**	**460**
Room-temperature performance	**E**	**E**	**E**	**E**	**E**	**E**	**E**
Edge definition	**G**	**G**	**E**	**E**	**E**	**E**	**E**
Dimensional stability	**G**		**G**	**G**	**E**	**E**	**E**
Hand carvable	**E**	**E**	**E**	**E**	**E**	**E**	**G**

good **E** excellent

Typical Properties	**5120**	**350**	**440**	**5440**	**450**	**5003**	**460**
Board Size, inches (size is approximate)	59×15.7×4	60x16x2 60x16x4	60×15.6×2 60×15.6×3 60×15.6×4	59x20x1 59x20x2 59x20x3 59x20x4	60x16x2 60x16x3 60x16x4 60x20x1 60x20x2 60x20x3 60x20x4	60x24x2 60x24x3 60x24x4 60x24x5 60x24x6	60x16x2 60x16x3 60x16x4 60x20x1 60x20x2 60x20x3 60x20x4

olor, Visual	Lt.Gray	Blue	Red Brn.	Lt. Brown	Lt. Brown	Tan	Lt. Brown
ensity, ASTM D-792,g/cc (lbs/ft^3)	0.25(15.5)	0.42(26)	0.55(34)	0.50(31)	0.66(41)	0.54(34)	0.77(48
ardness, ASTM D-2240, Shore D	28	30-35	54	55-60	65	60-64	64
lexural Strength, ASTM D-790, psi	700	750	2,300	2,520	4,000	3,400	3,700
lexural Modulus, ASTM D-790, psi	28,700	33,000	106,000		144,000	188,000	185,000
ltimate Tensile Strength, ASTM D-38, psi	500	550	1,300		2,300	2,900	1,800
ltimate Compressive Strength, STM D-695, psi	500	500	1,400	2,520	2,800	3,700	2,200
ompressive Modulus of Elasticity, STM D-695, PSI	28,900	38,000	105,000		135,000	185,000	181,000
lass Transition Temperature (Tg), MA, ASTM D-4065,°F(°C),E” Peak	208(98)	194(90)	203(95)		205(96)	182(83)	219(104
oefficient of Thermal Expansion er TMA, in/in/°F (-22°F to 86°F) n/in/°C (-30°C to 30°C)	33.6×10^{-6} 60.4×10^{-6}	41.7×10^{-6} 75×10^{-6}	34.4×10^{-6} 62×10^{-6}	$27\text{-}30\times10^{-6}$ $50\text{-}55\times10^{-6}$	36×10^{-6} 65×10^{-6}	32×10^{-6} 58×10^{-6}	29.9×10 53.8×10

oling Boards

pplications	**470**	**471**	**472**	**550**	**5005**	**5008**	**5010**	**5166**	**5168**	**5169**	**200**
ooling aids & fixtures	X	X	X					X	X	X	
aster Models				X	X	X		X			
utomotive die models						X					
ntermediate-temperature aerospace nasters & lay-up tools				X	X	X	X			X	
rototype foundry patterns	X	X	X	X				X	X	X	

oundry patterns							X	X			
rototype vacuum-form molds	X	X	X	X		X	X	X			X
etalforming tools							X	X	X		
njection molds									X		

good **E excellent**

omparison of Benefits	470	471	472	550	5005	5008	5010	5166	5168	5169	2000
oom-temperature performance	E	E	E	E	E	E	E	E	E	E	E
igh-temperature performance				G	G	G	E			E	
mpact resistance	E	G	G	G	G	G		E	E	E	E
brasionresistance	E	G	G	G		G		E	E	E	E
dge definition	E	E	E	E	E	E	E	E	E	E	E
imensional stability	E	E	E	E	E	E	E	E	E	E	E
and carvable					G	E					
hemical resistance	G			G	G	G	G	G	G	G	E

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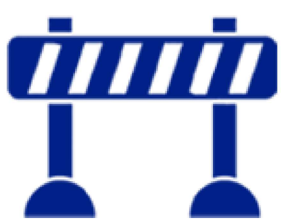

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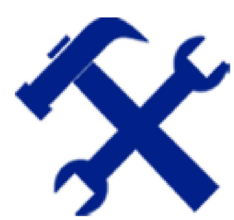

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[–] Location and Directions

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